4 Chemical potentials and quantum statistical physics – fermions and bosons

The problems are roughly in order of difficulty. Those indicated with ♡ are ones which are highly recommended. Any labelled with ♣ are the hardest ones, which might only occur as a "sting in the tail" at the end of a long examination question.

♡*Problem 4.1* **Chemical potentials and the Saha equation**
The notion that two systems which exchange particles must have equal chemical potentials in equilibrium may be used in discussing, as a function of temperature, the degree of ionisation of a gas of atomic hydrogen or the degree of dissociation of molecules into their constituent atoms.

Consider the case of the ionisation of a hydrogen gas. We will regard the density or temperature to be such that the bound hydrogen and the unbound protons and electrons behave as classical gases. We will make the assumption that we can neglect the excited states of bound hydrogen, so the atom is either in its ground state or ionised.

(i) Regarding the bound hydrogen gas as having chemical potential μ_{H} and the ionised gas as having chemical potential μ_{I}, what relation must they obey in equilibrium?

$$\mu_{\text{H}} = \mu_{\text{I}} \ .$$

We will regard the "Hydrogen universe" as electrically neutral. Consider the free energies

$$F_0(T, V, N_{\text{H}}, N_{\text{I}}) \quad \text{and} \quad F(T, V, N_{\text{H}}, N_{\text{p}}, N_{\text{e}}) \ ,$$

where N_{H}, N_{I}, N_{p} and N_{e} are the numbers of H atoms, ionised H atoms, protons and electrons in the universe.

(ii) Write down the expression for $\mathrm{d}F$ and deduce an expression for the chemical potentials, μ_{H}, μ_{p} and μ_{e} in terms of partial derivatives of F. In the following the constant variables are all the others apart from the one occurring in the partial derivative.

$$\begin{aligned} \mathrm{d}F &= \left(\frac{\partial F}{\partial T}\right)\mathrm{d}T + \left(\frac{\partial F}{\partial V}\right)\mathrm{d}V + \left(\frac{\partial F}{\partial N_{\text{H}}}\right)\mathrm{d}N_{\text{H}} + \left(\frac{\partial F}{\partial N_{\text{p}}}\right)\mathrm{d}N_{\text{p}} + \left(\frac{\partial F}{\partial N_{\text{e}}}\right)\mathrm{d}N_{\text{e}} \\ &= -S\mathrm{d}T - p\mathrm{d}V + \mu_{\text{H}}\mathrm{d}N_{\text{H}} + \mu_{\text{p}}\mathrm{d}N_{\text{p}} + \mu_{\text{e}}\mathrm{d}N_{\text{e}} \qquad (4.1) \\ \Rightarrow \quad \mu_j &= \frac{\partial F}{\partial N_j} \ , \end{aligned}$$

where j denotes which particle.
Assuming neutrality so that $\mathrm{d}N_{\text{p}} = \mathrm{d}N_{\text{e}} = \mathrm{d}N_{\text{I}}$, perform the same analysis with $\mathrm{d}F_0$ and its partial derivatives and deduce a relation between μ_{p} and μ_{e}, and μ_{I}.

$$\begin{aligned} \mathrm{d}F_0 &= \left(\frac{\partial F}{\partial V}\right)\mathrm{d}V + \left(\frac{\partial F}{\partial N_{\text{H}}}\right)\mathrm{d}N_{\text{H}} + \left(\frac{\partial F}{\partial N_{\text{I}}}\right)\mathrm{d}N_{\text{I}} \\ &= -S\mathrm{d}T - p\mathrm{d}V + \mu_{\text{H}}\mathrm{d}N_{\text{H}} + \mu_{\text{I}}\mathrm{d}N_{\text{I}} \end{aligned}$$

(Use Eq. (4.1):) $\quad \Rightarrow \quad \mu_{\text{I}} = \mu_{\text{p}} + \mu_{\text{e}} \ ,$

using $\mathrm{d}N_{\mathsf{p}} = \mathrm{d}N_{\mathsf{e}} = \mathrm{d}N_{\mathrm{I}}$.

Now consider the simplified Hamiltonian for the H atom

$$\mathcal{H} = \frac{p^2}{2m_{\mathrm{H}}} - \varepsilon_0 \ ,$$

where m_{H} is the mass of the H atom and ε_0 is the binding energy.

(iii) Write down the partition function for a (bound) H atom and hence its free energy.

$$\begin{aligned} Z(T,V,N_{\mathrm{H}}=1) &= \frac{\int \mathrm{d}V \int \mathrm{d}^3\mathbf{p}}{\mathsf{h}^3} \exp\left\{-\beta\left(\frac{p^2}{2m_{\mathrm{H}}} - \varepsilon_0\right)\right\} \\ &= \underbrace{\frac{1}{\mathsf{h}^3}\int \mathrm{d}V \left[\int \mathrm{d}^3\mathbf{p} \exp\left\{-\beta\left(\frac{p^2}{2m_{\mathrm{H}}}\right)\right\}\right]}_{\text{Ideal gas } Z} \mathrm{e}^{\beta\varepsilon_0} \end{aligned} \tag{4.2}$$

$$\Rightarrow \quad F(T,V,N_{\mathrm{H}}=1) = -k_{\mathrm{B}}T \ln Z(T,V,N_{\mathrm{H}}=1) \ .$$

(iv) Write down the free energy for a classical ideal gas composed of N_{H} such H atoms (remembering indistinguishability) and hence deduce the chemical potential for an H atom.

The only difference from the classical ideal gas is the term $\mathrm{e}^{\beta\varepsilon_0}$ in Eq. (4.2), the Boltzmann factor for the binding energy, so we find (in an exam you would be expected to derive this, including use of Stirling approximation for $N!$):

$$F(T,V,N_{\mathrm{H}}) = -Nk_{\mathrm{B}}T\left\{\ln\left(\frac{v_{\mathrm{H}}}{\lambda_{\mathrm{TH}}^3}\mathrm{e}^{\beta\varepsilon_0}\right) + 1\right\} \text{ , where } v_{\mathrm{H}} = \frac{V}{N_{\mathrm{H}}} \text{ and } \lambda_{\mathrm{TH}} = \left(\frac{\mathsf{h}^2}{2\pi m_{\mathrm{H}} k_{\mathrm{B}} T}\right)^{1/2}$$

and hence the chemical potential is

$$\begin{aligned} \mu_{\mathrm{H}} &= -k_{\mathrm{B}}T \ln\left(\frac{v_{\mathrm{H}}}{\lambda_{\mathrm{TH}}^3}\mathrm{e}^{\beta\varepsilon_0}\right) \\ &= -\varepsilon_0 - k_{\mathrm{B}}T \ln\left(\frac{v_{\mathrm{H}}}{\lambda_{\mathrm{TH}}^3}\right) \ . \end{aligned} \tag{4.3}$$

Note the first term is the internal energy of the bound H atom. The remainder comes from the centre of mass motion.

(v) Regarding the atoms, protons and electrons as ideal classical gases of identical particles use the result of the previous part of this question and the relationship between the chemical potentials derived in part (ii) to show

$$\lambda_{\mathrm{Te}}^3 \lambda_{\mathrm{Tp}}^3 \, n_{\mathsf{p}} n_{\mathsf{e}} = \lambda_{\mathrm{TH}}^3 \, n_{\mathrm{H}} \mathrm{e}^{-\beta\varepsilon_0} \ ,$$

where for particle species j

$$n_j = \frac{N_j}{V} \quad \text{and} \quad \lambda_{\mathrm{T}j} = \left(\frac{\mathsf{h}^2}{2\pi m_j k_{\mathrm{B}} T}\right)^{1/2} \ .$$

We have the equality of the chemical potentials

$$\begin{aligned} \mu_{\text{H}} &= \mu_{\text{I}} \\ &= \mu_{\text{p}} + \mu_{\text{e}} \\ \Rightarrow \quad \ln\left(\frac{v_{\text{H}}}{\lambda_{\text{TH}}^3}\mathrm{e}^{\beta\varepsilon_0}\right) &= \ln\left(\frac{v_{\text{e}}}{\lambda_{\text{Te}}^3}\right) + \ln\left(\frac{v_{\text{p}}}{\lambda_{\text{Tp}}^3}\right) \end{aligned}$$

So exponentiating and inverting we find

$$\lambda_{\text{Te}}^3\lambda_{\text{Tp}}^3\, n_{\text{p}}n_{\text{e}} = \lambda_{\text{TH}}^3\, n_{\text{H}}\mathrm{e}^{-\beta\varepsilon_0} \ .$$

(vi) Recognising the neutrality of the gas and the disparity in mass between the electrons and protons show (a form of Saha's equation)

$$n - n_{\text{I}} = \mathrm{e}^{\beta\varepsilon_0}\lambda_{\text{Te}}^3 n_{\text{I}}^2 \ ,$$

where $n = N/V$ and N is the total number of protons – free or bound in H atoms. We see that since $m_{\text{e}} \ll m_{\text{p}}$, $m_{\text{H}} \simeq m_{\text{p}}$, so the thermal de Broglie wavelength factors cancel. Also because of neutrality, $n_{\text{p}} = n_{\text{e}} = n_{\text{I}}$. Then writing $n_{\text{H}} = n - n_{\text{I}}$ we find result:

$$n - n_{\text{I}} = \mathrm{e}^{\beta\varepsilon_0}\lambda_{\text{Te}}^3 n_{\text{I}}^2 \ . \tag{4.4}$$

(vii) Hence show that

$$n_{\text{I}} = \frac{-1 \pm \sqrt{1 + 4\mathrm{e}^{\beta\varepsilon_0}\lambda_{\text{Te}}^3 n}}{2\mathrm{e}^{\beta\varepsilon_0}\lambda_{\text{Te}}^3} \ , \tag{4.5}$$

choosing the appropriate sign.

Note Eq. (4.4) is a quadratic in n_{I}, so we may write down the above result. The number density, $n_{\text{I}} \geq 0$, so we must pick the positive sign.

(viii) Deduce the expressions for n_{I} as $T \to 0$ and $T \to \infty$. Comment on the behaviour of $n_{\text{I}}(T, V, N)$ in the limit that $V \to \infty$ with N and T fixed.

From Eq. (4.5) we note as $T \to 0$, or $\beta \to \infty \Rightarrow \mathrm{e}^{\beta\varepsilon_0} \to \infty$, so we find that

$$\begin{aligned} n_{\text{I}} &= \frac{-1 \pm \sqrt{1 + 4\mathrm{e}^{\beta\varepsilon_0}\lambda_{\text{Te}}^3 n}}{2\mathrm{e}^{\beta\varepsilon_0}\lambda_{\text{Te}}^3} \\ &\simeq \sqrt{n\lambda_{\text{Te}}^{-3}}\mathrm{e}^{-\beta\varepsilon_0/2} \ . \end{aligned}$$

When $T \to \infty$, $\mathrm{e}^{\beta\varepsilon_0} \to 1$, and $\lambda_{\text{Te}} \to 0$, so the second term inside the square root in Eq. (4.5) tends to zero and we may expand. We find

$$\begin{aligned} n_{\text{I}} &= \frac{-1 \pm \sqrt{1 + 4\mathrm{e}^{\beta\varepsilon_0}\lambda_{\text{Te}}^3 n}}{2\mathrm{e}^{\beta\varepsilon_0}\lambda_{\text{Te}}^3} \\ &\simeq \frac{-1 + 1 + \frac{1}{2}4\mathrm{e}^{\beta\varepsilon_0}\lambda_{\text{Te}}^3 n}{2\mathrm{e}^{\beta\varepsilon_0}\lambda_{\text{Te}}^3} \\ &= n \ . \end{aligned}$$

I.e. all atoms are ionised at high temperatures.

Finally note that if we fix T and N, but let $V \to \infty$ (a large but rather empty universe), again the second term in the square root tends to zero, from the factor of $n = N/V$, and so we obtain all atoms ionised *even if T is low*. This is caused by the positional entropy, gained by separating the electrons and protons, dominating at large volumes.

Problem 4.2 **The Yang-Yang equation**
To see that

$$C_{V,N} = T\left(\frac{\partial S}{\partial T}\right)_{V,N} ,$$

Consider the First law for reversible changes,

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V + \mu\mathrm{d}N \quad \text{and} \quad đQ = T\mathrm{d}S \ .$$

Then letting S be a function of T, V and N, we see

$$\mathrm{d}S = \left(\frac{\partial S}{\partial T}\right)_{V,N}\mathrm{d}T + \left(\frac{\partial S}{\partial V}\right)_{T,N}\mathrm{d}V + \left(\frac{\partial S}{\partial N}\right)_{V,T}\mathrm{d}N \ .$$

So at constant volume and number

$$đQ = C_{V,N}\mathrm{d}T = T\left(\frac{\partial S}{\partial T}\right)_{V,N}\mathrm{d}T \quad \Rightarrow \quad C_{V,N} = T\left(\frac{\partial S}{\partial T}\right)_{V,N} \ .$$

From the Gibbs-Duhem relation,

$$-S\mathrm{d}T - N\mathrm{d}\mu + V\mathrm{d}p = 0 \tag{4.6}$$

we may regard μ and p as functions of T, V and N

$$\mathrm{d}\mu = \left(\frac{\partial \mu}{\partial T}\right)_{V,N}\mathrm{d}T + \left(\frac{\partial \mu}{\partial V}\right)_{T,N}\mathrm{d}V + \left(\frac{\partial \mu}{\partial N}\right)_{T,V}\mathrm{d}N$$
$$\mathrm{d}p = \left(\frac{\partial p}{\partial T}\right)_{V,N}\mathrm{d}T + \left(\frac{\partial p}{\partial V}\right)_{T,N}\mathrm{d}V + \left(\frac{\partial p}{\partial N}\right)_{T,V}\mathrm{d}N \ .$$

Substituting these relations into Eq. (4.6) **at constant volume and number** we see that

$$S = -N\left(\frac{\partial \mu}{\partial T}\right)_{V,N} + V\left(\frac{\partial p}{\partial T}\right)_{V,N}$$
$$\Rightarrow \quad T\left(\frac{\partial}{\partial T}\right)_{V,N} S = T\left(\frac{\partial}{\partial T}\right)_{V,N}\left[-N\left(\frac{\partial \mu}{\partial T}\right)_{V,N} + V\left(\frac{\partial p}{\partial T}\right)_{V,N}\right]$$
$$\Rightarrow \quad C_{V,N} = -NT\left(\frac{\partial^2 \mu}{\partial T^2}\right)_{V,N} + VT\left(\frac{\partial^2 p}{\partial T^2}\right)_{V,N} ,$$

the *Yang-Yang* equation.

♡*Problem 4.3* **Neutrinos**
We have a linear dispersion law. For linear dispersion with $\varepsilon_{\mathbf{k}} = cp = \hbar ck$. Ignoring spin

there is one quantum state associated with a "volume" $(2\pi)^3/V$ of k-space and so in a spherical shell of volume $4\pi k^2 \mathrm{d}k$ there are

$$\frac{4\pi k^2 \mathrm{d}k}{(2\pi)^3/V} = \frac{V}{(2\pi)^3} 4\pi k^2 \mathrm{d}k$$

quantum states. The density of states in energy is defined by

$$g(\varepsilon)\,\mathrm{d}\varepsilon = \frac{V}{(2\pi)^3} 4\pi k^2 \mathrm{d}k \Longrightarrow g(\varepsilon) = \frac{V}{(2\pi)^3} 4\pi k^2 \frac{\mathrm{d}k}{\mathrm{d}\varepsilon}$$

For a linear dispersion law $\mathrm{d}\varepsilon/\mathrm{d}k = \hbar \mathrm{c}$ and so

$$g(\varepsilon) = \frac{V}{(2\pi)^3} \frac{4\pi k^2}{\hbar \mathrm{c}} = \frac{V}{2\pi^2 \hbar \mathrm{c}} k^2 = \frac{V}{2\pi^2 (\hbar \mathrm{c})^3} \varepsilon^2 = \frac{V 4\pi}{h^3 \mathrm{c}^3} \varepsilon^2.$$

We write this as $g(\varepsilon) = A\varepsilon^2$, for convenience, with

$$A = \frac{V 4\pi}{h^3 \mathrm{c}^3}$$

. Because the neutrinos are "chiral" Fermions there is only 1 neutrino in every single particle wave state and so in converting to an integral over energies we must write

$$\sum_{i \text{ states}} \cdots \to \int_0^\infty \mathrm{d}\varepsilon\; g(\varepsilon) \cdots$$

The Fermi occupancy factor gives the average number of fermions in a state with energy $g(\varepsilon)$ and is

$$\bar{n}(\varepsilon) = \frac{1}{\mathrm{e}^{(\varepsilon-\mu)/k_\mathrm{B}T} + 1}.$$

We are told that the neutrinos are not conserved and so their mean number in thermal equilibrium at constant volume and temperature is determined by minimising the Helmholtz Free Energy i.e.

$$\frac{\partial F}{\partial \bar{N}} = 0 = \mu.$$

Thus as for non-conserved photons in black body radiation the chemical potential $\mu = 0$ and so the Fermi Function becomes

$$\bar{n}(\varepsilon) = \frac{1}{\mathrm{e}^{\varepsilon/k_\mathrm{B}T} + 1}$$

The internal, or mean, energy is thus

$$\begin{aligned} U &= \int_0^\infty \mathrm{d}\varepsilon\; g(\varepsilon)\; \varepsilon\, \bar{n}(\varepsilon) \\ &= \int_0^\infty \mathrm{d}\varepsilon\, A\varepsilon^2 \varepsilon\, \frac{1}{\mathrm{e}^{\varepsilon/k_\mathrm{B}T} + 1} \\ &= A \int_0^\infty \mathrm{d}\varepsilon\; \frac{\varepsilon^3}{\mathrm{e}^{\varepsilon/k_\mathrm{B}T} + 1}. \end{aligned}$$

Change to the dimensionless variable $z = \varepsilon/k_{\text{B}}T$ so that $\varepsilon = k_{\text{B}}Tz$ and $\mathrm{d}\varepsilon = k_{\text{B}}T\ \mathrm{d}z$. Then

$$U = A\,(k_{\text{B}}T)^4 \int_0^\infty \mathrm{d}z\ \frac{z^3}{\mathrm{e}^z + 1}$$

We are told that

$$\int_0^\infty \mathrm{d}z\ \frac{z^3}{\mathrm{e}^z + 1} = \frac{7\pi^4}{120}$$

and so

$$U = A\,(k_{\text{B}}T)^4\,\frac{7\pi^4}{120} = \frac{V4\pi}{h^3\mathrm{c}^3}\frac{7\pi^4}{120}\,(k_{\text{B}}T)^4 = \frac{V4\pi}{(2\pi)^3\,\hbar^3\mathrm{c}^3}\frac{7\pi^4}{120}\,(k_{\text{B}}T)^4 = \frac{V}{\hbar^3\mathrm{c}^3}\frac{7\pi^2}{240}\,(k_{\text{B}}T)^4$$

as required.

For a system with a linear dispersion law, $pV = U/3$ and so

$$p = \frac{1}{\hbar^3\mathrm{c}^3}\frac{7\pi^2}{720}\,(k_{\text{B}}T)^4$$

♡*Problem 4.4* **Ripplons**

In a one dimensional wave problem in a box of length L there is one quantum state in an interval $\Delta k_x = 2\pi/L$. In two dimensions in a square box of area $L^2 = A$ there will thus be one quantum state in an area $(2\pi/L)^2 = (2\pi)^2/A$ of k-space. In a problem in which ε depends on $k = |\mathbf{k}|$ all points the same distance from the origin in k-space have the same energy and so the number of states in a thin annulus of width $\mathrm{d}k$ is $2\pi k\,\mathrm{d}k/\left((2\pi)^2/A\right)$. All these states have energies between ε and $\varepsilon + \mathrm{d}\varepsilon$ and so the density of states $g(\varepsilon)$ is

$$g(\varepsilon)\mathrm{d}\varepsilon = \frac{A}{(2\pi)^2}2\pi k\,\mathrm{d}k \quad \Rightarrow \quad g(\varepsilon) = \frac{A}{(2\pi)^2}\left(\frac{2\pi k}{\mathrm{d}\varepsilon/\mathrm{d}k}\right)_{\varepsilon=\varepsilon_k}.$$

In this example $\varepsilon_k = Dk^{3/2}$ and so $k^{1/2} = (\varepsilon_k/D)^{1/3}$

$$\begin{aligned} g(\varepsilon) &= \frac{A}{(2\pi)^2}\left(\frac{2}{3}\frac{2\pi k}{Dk^{1/2}}\right)_{\varepsilon=\varepsilon_k} \\ &= \frac{A}{(2\pi)^2}\left(\frac{2}{3}\frac{2\pi}{D}\frac{\varepsilon^{1/3}}{D^{1/3}}\right) \\ &= \frac{A}{3\pi D^{4/3}}\,\varepsilon^{1/3}. \end{aligned}$$

(i) $F = U - TS$ and

$$\begin{aligned} \mathrm{d}F &= \mathrm{d}U - T\mathrm{d}S - S\mathrm{d}T \\ &= T\mathrm{d}S - p\,\mathrm{d}V + \mu\mathrm{d}N - T\mathrm{d}S - S\mathrm{d}T \\ &= -p\,\mathrm{d}V + \mu\mathrm{d}N - S\mathrm{d}T \end{aligned}$$

and so

$$\mu = \left(\frac{\partial F}{\partial N}\right)_{T,V}.$$

If the boson-like excitations are not conserved then equilibrium in particle number is determined by minimising the Free Energy as a function of N and so $(\partial F/\partial N)_{T,V} = 0 = \mu$ as required.

(ii) We can use the mean occupation numbers for Bosons with $\mu = 0$ and so

$$\langle n_k \rangle = \frac{1}{\mathrm{e}^{\beta\varepsilon_k} - 1}$$

and
$$\langle N \rangle = \int_0^\infty \mathrm{d}\varepsilon\; g(\varepsilon)\,\langle n(E)\rangle = \frac{A}{3\pi D^{4/3}} \int_0^\infty \mathrm{d}\varepsilon\, \varepsilon^{1/3}\, \frac{1}{\mathrm{e}^{\varepsilon/k_\mathrm{B}T} - 1}.$$

Write $z = \varepsilon/k_\mathrm{B}T$ and $\mathrm{d}\varepsilon = k_\mathrm{B}T\,\mathrm{d}z$ then

$$\langle N \rangle = \frac{A}{3\pi D^{4/3}} (k_\mathrm{B}T)^{4/3} \underbrace{\int_0^\infty \mathrm{d}z\, \frac{z^{1/3}}{\mathrm{e}^z - 1}}_{=I} = (k_\mathrm{B}T)^{4/3} \left(\frac{A}{3\pi D^{4/3}}\right) I\ ,$$

as required.

(iii) The mean internal energy of the system is

$$\begin{aligned} U &= \int_0^\infty \mathrm{d}\varepsilon\; g(\varepsilon)\varepsilon\,\langle n(\varepsilon)\rangle \\ &= \frac{A}{3\pi D^{4/3}} \int_0^\infty \mathrm{d}\varepsilon\, \varepsilon^{4/3}\, \frac{1}{\mathrm{e}^{\varepsilon/k_\mathrm{B}T} - 1} \\ &= \frac{A}{3\pi D^{4/3}} (k_\mathrm{B}T)^{7/3} \underbrace{\int_0^\infty \mathrm{d}z\, \frac{z^{4/3}}{\mathrm{e}^z - 1}}_{=J} \\ &= (k_\mathrm{B}T)^{7/3} \left(\frac{A}{3\pi D^{4/3}}\right) J\ . \end{aligned}$$

(iv)

$$C_A = \left(\frac{\partial U}{\partial T}\right)_A = k_\mathrm{B}^{7/3} \left(\frac{7T^{4/3}}{3}\right) \left(\frac{A}{3\pi D^{4/3}}\right) J = k_\mathrm{B}\,(k_\mathrm{B}T)^{4/3} \left(\frac{7A}{9\pi D^{4/3}}\right) J.$$

(v) Third law: $S = k_\mathrm{B} \ln \Omega$. Expect that as $T \to 0$ system will end up in its ground state and all known ground states are non degenerate i.e. $\Omega = 1$ and so $S \to 0$ as $T \to 0$. Now

$$\mathrm{d}U = \underbrace{T\mathrm{d}S}_{\text{Heat term}} + \underbrace{\gamma \mathrm{d}A}_{\text{work term}}\ ,$$

where γ is the surface tension.
Thus at constant area $\mathrm{d}U = T\mathrm{d}S = C_A \mathrm{d}T$ and

$$C_A = T\left(\frac{\partial S}{\partial T}\right)_A \quad \Rightarrow \quad S(T) - S(T_0) = \int_{T_0}^T \mathrm{d}T' \frac{C_A}{T'}.$$

We have seen that $C_A \propto T^{4/3}$ and so $\to 0$ as $T \to 0$. Thus choose $T_0 = 0$ and

$S(0) \to 0$ and so

$$\begin{aligned} S(T) &= \int_0^T \mathrm{d}T' \frac{C_A}{T'} \\ &= \left(\frac{7A}{9\pi D^{4/3}}\right) J k_{\mathrm{B}}^{7/3} \int_0^T \mathrm{d}T' \, \frac{T'^{4/3}}{T'} \\ &= \left(\frac{7A}{9\pi D^{4/3}}\right) J k_{\mathrm{B}}^{7/3} \frac{T^{4/3}}{4/3} \\ &= \frac{3}{4}\left(\frac{7A}{9\pi D^{4/3}}\right) J k_{\mathrm{B}} (k_{\mathrm{B}} T)^{4/3} \end{aligned}$$

(vi)

$$\begin{aligned} F = U - TS &= U - \frac{3}{4}\left(\frac{7A}{9\pi D^{4/3}}\right) J (k_{\mathrm{B}} T)^{7/3} \\ &= U - \frac{7}{4}\underbrace{\left(\frac{A}{3\pi D^{4/3}}\right) J (k_{\mathrm{B}} T)^{7/3}}_{=U} \\ &= U - \frac{7}{4} U \\ &= -\frac{3}{4} U \ , \end{aligned}$$

as required.

Problem 4.5 **Entropy and fermion occupation numbers**
The fermion occupation number of state $\mathbf{k}\sigma$, $n_{\mathbf{k}\sigma}$, are

$$n_{\mathbf{k}\sigma} = \frac{1}{\mathrm{e}^{\beta(\varepsilon_{\mathbf{k}\sigma} - \mu)} + 1} \ . \tag{4.7}$$

(i) Express $\mathrm{e}^{\beta(\varepsilon_{\mathbf{k}\sigma} - \mu)}$ in terms of $n_{\mathbf{k}\sigma}$.

$$\begin{aligned} \mathrm{e}^{\beta(\varepsilon_{\mathbf{k}\sigma} - \mu)} &= \frac{1}{n_{\mathbf{k}\sigma}} - 1 \\ &= \frac{1 - n_{\mathbf{k}\sigma}}{n_{\mathbf{k}\sigma}} \ . \end{aligned} \tag{4.8}$$

(ii) Express $1 + \mathrm{e}^{-\beta(\varepsilon_{\mathbf{k}\sigma} - \mu)}$ (noting the minus sign in the exponent) in terms of $n_{\mathbf{k}\sigma}$.

$$\begin{aligned} 1 + \mathrm{e}^{-\beta(\varepsilon_{\mathbf{k}\sigma} - \mu)} &= 1 + \frac{n_{\mathbf{k}\sigma}}{1 - n_{\mathbf{k}\sigma}} \\ &= \frac{1}{1 - n_{\mathbf{k}\sigma}} \ . \end{aligned} \tag{4.9}$$

The Grand energy, Ω, for fermions is

$$\Omega = -k_{\mathrm{B}} T \sum_{\mathbf{k}\sigma} \ln\left(1 + \mathrm{e}^{-\beta(\varepsilon_{\mathbf{k}\sigma} - \mu)}\right) \ . \tag{4.10}$$

(iii) Use Eq. (4.10):

$$
\begin{aligned}
S &= -\left(\frac{\partial\Omega}{\partial T}\right)_{V,\mu} \\
&= k_{\mathrm{B}}\sum_{\mathbf{k}\sigma}\ln\left(1+\mathrm{e}^{-\beta(\varepsilon_{\mathbf{k}\sigma}-\mu)}\right) + k_{\mathrm{B}}T\frac{\partial\beta}{\partial T}\sum_{\mathbf{k}\sigma}\frac{\partial}{\partial\beta}\ln\left(1+\mathrm{e}^{-\beta(\varepsilon_{\mathbf{k}\sigma}-\mu)}\right) \\
&= k_{\mathrm{B}}\sum_{\mathbf{k}\sigma}\left[\ln\left(1+\mathrm{e}^{-\beta(\varepsilon_{\mathbf{k}\sigma}-\mu)}\right) + T\left(-\frac{1}{k_{\mathrm{B}}T^2}\right)(-1)\frac{(\varepsilon_{\mathbf{k}\sigma}-\mu)\mathrm{e}^{-\beta(\varepsilon_{\mathbf{k}\sigma}-\mu)}}{1+\mathrm{e}^{-\beta(\varepsilon_{\mathbf{k}\sigma}-\mu)}}\right] \\
&= k_{\mathrm{B}}\sum_{\mathbf{k}\sigma}\left[\ln\left(1+\mathrm{e}^{-\beta(\varepsilon_{\mathbf{k}\sigma}-\mu)}\right) + \frac{\beta(\varepsilon_{\mathbf{k}\sigma}-\mu)}{1+\mathrm{e}^{\beta(\varepsilon_{\mathbf{k}\sigma}-\mu)}}\right] \\
&= k_{\mathrm{B}}\sum_{\mathbf{k}\sigma}\left[\ln\left(1+\mathrm{e}^{-\beta(\varepsilon_{\mathbf{k}\sigma}-\mu)}\right) + \ln\left(\mathrm{e}^{\beta(\varepsilon_{\mathbf{k}\sigma}-\mu)}\right)\, n_{\mathbf{k}\sigma}\right] \\
&= k_{\mathrm{B}}\sum_{\mathbf{k}\sigma}\left\{-\ln(1-n_{\mathbf{k}\sigma}) + n_{\mathbf{k}\sigma}\left[\ln(1-n_{\mathbf{k}\sigma}) - \ln n_{\mathbf{k}\sigma}\right]\right\} \quad \text{(Using Eq. (4.8), Eq. (4.9).)} \\
&= -k_{\mathrm{B}}\sum_{\mathbf{k}\sigma}\left\{n_{\mathbf{k}\sigma}\ln n_{\mathbf{k}\sigma} + (1-n_{\mathbf{k}\sigma})\ln(1-n_{\mathbf{k}\sigma})\right\}\ , \qquad (4.11)
\end{aligned}
$$

using the identities established in the the first two parts of this problem.

(iv) Using $\Omega = U - TS - \mu N$, determine S in terms of $n_{\mathbf{k}\sigma}$ and show consistency with the last part of the question.

$$
\begin{aligned}
TS &= U - \Omega - \mu N \\
\Rightarrow \quad S &= k_{\mathrm{B}}\beta\left(U - \Omega - \mu N\right) \\
&= k_{\mathrm{B}}\beta\sum_{\mathbf{k}\sigma}\left[\varepsilon_{\mathbf{k}\sigma}n_{\mathbf{k}\sigma} - (-k_{\mathrm{B}}T)\ln\left(1+\mathrm{e}^{-\beta(\varepsilon_{\mathbf{k}\sigma}-\mu)}\right) - \mu n_{\mathbf{k}\sigma}\right] \\
&= k_{\mathrm{B}}\sum_{\mathbf{k}\sigma}\left[n_{\mathbf{k}\sigma}\beta(\varepsilon_{\mathbf{k}\sigma}-\mu) + \ln\left(1+\mathrm{e}^{-\beta(\varepsilon_{\mathbf{k}\sigma}-\mu)}\right)\right] \\
&= k_{\mathrm{B}}\sum_{\mathbf{k}\sigma}\left[n_{\mathbf{k}\sigma}\ln\left(\beta(\varepsilon_{\mathbf{k}\sigma}-\mu)\right) + \ln\left(1+\mathrm{e}^{-\beta(\varepsilon_{\mathbf{k}\sigma}-\mu)}\right)\right] \\
&= k_{\mathrm{B}}\sum_{\mathbf{k}\sigma}\left[n_{\mathbf{k}\sigma}\ln\left(\mathrm{e}^{\beta(\varepsilon_{\mathbf{k}\sigma}-\mu)}\right) + \ln\left(1+\mathrm{e}^{-\beta(\varepsilon_{\mathbf{k}\sigma}-\mu)}\right)\right] \\
&= k_{\mathrm{B}}\sum_{\mathbf{k}\sigma}\left[n_{\mathbf{k}\sigma}\ln\left(\frac{1-n_{\mathbf{k}\sigma}}{n_{\mathbf{k}\sigma}}\right) + \ln\left(\frac{1}{1-n_{\mathbf{k}\sigma}}\right)\right] \quad \text{(Using Eq. (4.8), Eq. (4.9).)} \\
&= -k_{\mathrm{B}}\sum_{\mathbf{k}\sigma}\left\{n_{\mathbf{k}\sigma}\ln n_{\mathbf{k}\sigma} + (1-n_{\mathbf{k}\sigma})\ln(1-n_{\mathbf{k}\sigma})\right\}\ ,
\end{aligned}
$$

consistent with Eq. (4.11).

(v) Take differentials of both sides of Eq. (4.11) and deduce that if all occupation numbers are kept constant S does not change.

This is obvious as on the righthand side the only variable is $n_{\mathbf{k}\sigma}$, so there will be a common factor, inside the sum, of $\mathrm{d}n_{\mathbf{k}\sigma}$.

Problem 4.6 **Pauli magnetic susceptibility**

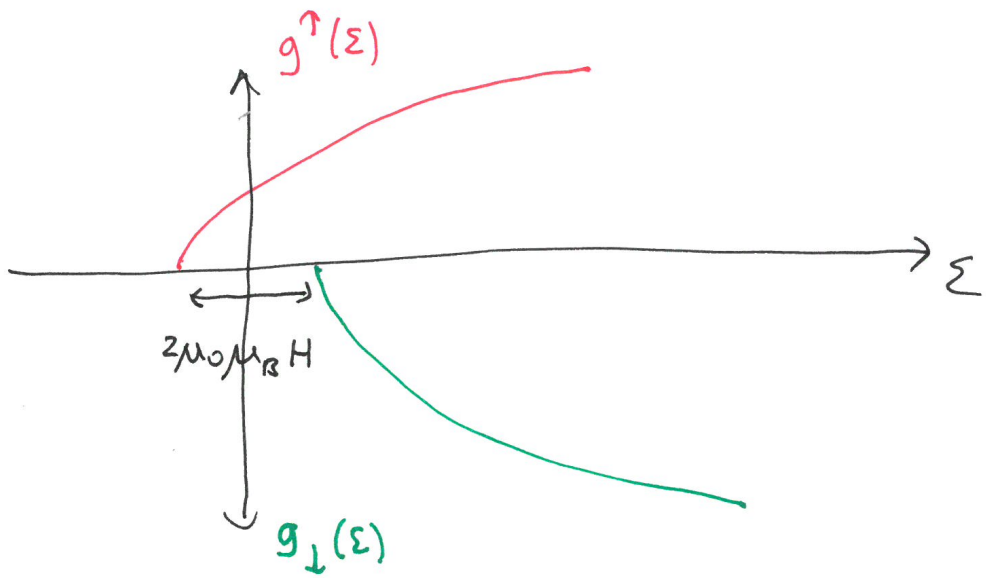

Figure 4.1: The density of states for up-spin, $g^{\uparrow}(\varepsilon)$ and down-spin $g^{\downarrow}(\varepsilon)$ particles.

(i) The densities of states, for three dimensions, are sketched in Fig. (4.1).

(ii) Let

$$N^{\uparrow} = \tfrac{1}{2}N + \Delta \quad \text{and} \quad N^{\downarrow} = \tfrac{1}{2}N - \Delta \ ,$$

ensuring that $N = N^{\uparrow} + N^{\downarrow}$. The the total energy, $U(N, \Delta, H)$, is (noting that for each of the up- and down-spin electrons, the "$2S+1$" $= 1$, as there is only one spin state.)

$$\begin{aligned} U(N,\Delta,H) &= -\left(\tfrac{1}{2}N + \Delta\right)\mu_0\mu_{\text{B}}H + \frac{3}{5}\underbrace{(6\pi^2)}_{\text{not } 3\pi^2}{}^{2/3}\frac{\hbar^2}{2m}\frac{\left(\frac{1}{2}N+\Delta\right)^{5/3}}{V^{2/3}} + \\ &\quad + \left(\tfrac{1}{2}N - \Delta\right)\mu_0\mu_{\text{B}}H + \frac{3}{5}(6\pi^2)^{2/3}\frac{\hbar^2}{2m}\frac{\left(\frac{1}{2}N-\Delta\right)^{5/3}}{V^{2/3}} \\ &= -2\Delta\mu_0\mu_{\text{B}}H + \frac{3}{5}(6\pi^2)^{2/3}\frac{\hbar^2}{2mV^{2/3}}\left(\left[\tfrac{1}{2}N+\Delta\right]^{5/3} + \left[\tfrac{1}{2}N-\Delta\right]^{5/3}\right) \end{aligned}$$

Now we look for the stationary point with respect to Δ, i.e. the equilibrium magneti-

sation,

$$\begin{aligned}\frac{\partial U}{\partial \Delta} &= -2\mu_0\mu_{\rm B}H + \frac{3}{5}(6\pi^2)^{2/3}\frac{\hbar^2}{2mV^{2/3}}\frac{5}{3}\left(\left[\tfrac{1}{2}N+\Delta\right]^{2/3} - \left[\tfrac{1}{2}N-\Delta\right]^{2/3}\right)\\ &= -2\mu_0\mu_{\rm B}H + (6\pi^2)^{2/3}\frac{\hbar^2}{2mV^{2/3}}\left\{\left(\frac{N}{2}\right)^{2/3}\left[1+\frac{2\Delta}{N}\right]^{2/3} - \left[1-\frac{2\Delta}{N}\right]^{2/3}\right\}\end{aligned} \tag{4.12}$$

$$\begin{aligned}\text{(Use } \Delta/N \ll 1\text{:)}\quad &\simeq -2\mu_0\mu_{\rm B}H + (6\pi^2)^{2/3}\frac{\hbar^2}{2mV^{2/3}}\left\{\left(\frac{N}{2}\right)^{2/3}\left[1+\frac{4\Delta}{3N} - 1 + \frac{4\Delta}{3N}\right]\right\}\\ &= -2\mu_0\mu_{\rm B}H + \frac{(6\pi^2)^{2/3}}{2^{2/3}}\frac{\hbar^2 n^{2/3}}{2m}\frac{8\Delta}{3N}\\ &= -2\mu_0\mu_{\rm B}H + \underbrace{(3\pi^2)^{2/3}\frac{\hbar^2 n^{2/3}}{2m}}_{=\varepsilon_{\rm F}(H=0)}\frac{8\Delta}{3N}\\ &= -2\mu_0\mu_{\rm B}H + \varepsilon_{\rm F}(H=0)\frac{8\Delta}{3N}\\ \Rightarrow\quad \Delta &= \frac{3}{4}\frac{\mu_o\mu_{\rm B}H}{\varepsilon_{\rm F}(H=0)}N\ .\end{aligned}$$

(iii) The magnetisation is:

$$\begin{aligned}M &= \left[\left(\tfrac{1}{2}N+\Delta\right) - \left(\tfrac{1}{2}N-\Delta\right)\right]\mu_{\rm B}\\ &= 2\Delta\mu_{\rm B}\\ &= N\mu_{\rm B}\frac{3}{2}\frac{\mu_0\mu_{\rm B}H}{\varepsilon_{\rm F}(H=0)}\end{aligned}$$

This expression has a physical interpretation. If the magnetic moments were classical, all would be aligned at $T=0$, so $M_{\text{classical}} = N\mu_{\rm B}$. the fermi gas result is because the up and down spins in the frozen part of the fermi sea ($\varepsilon < \varepsilon_{\rm F}(H=0) - \mu_0\mu_{\rm B}H$) cancel and it is only in the region in energy of order $2\mu_0\mu_{\rm B}H$ where the up-spins predominate. Thus the fraction of spins forming the magnetisation is suppressed by the ratio in the expression.

(iv) The Pauli susceptibility, χ_{Pauli} is

$$\chi_{\text{Pauli}} = \frac{\partial M}{\partial H} = N\mu_{\rm B}\frac{3}{2}\frac{\mu_0\mu_{\rm B}}{\varepsilon_{\rm F}(H=0)}\ .$$

(v) To determine the field which polarises the gas, note that the number of down-spins must be zero. So $\Delta = N/2$, and using that in Eq. (4.12), we see

$$0 = -2\mu_0\mu_{\rm B}H + (6\pi^2)^{2/3}\frac{\hbar^2}{2mV^{2/3}}\left\{\left(\frac{N}{2}\right)^{2/3}\Big[\underbrace{1+\frac{2\Delta}{N}}_{=2}\Big]^{2/3} - \Big[\underbrace{1-\frac{2\Delta}{N}}_{=0}\Big]^{2/3}\right\}$$

$$\begin{aligned}\Rightarrow\quad 2\mu_0\mu_{\rm B}H &= (6\pi^2)^{2/3}\frac{\hbar^2 n^{2/3}}{2m}\\ &= \varepsilon_{\rm F}(H=0)\end{aligned}$$

This has the interpretation of displacing the two densities of states so that the single spin fermi energy lies (just) below the start of the down-spin density of states – see Fig. (4.1).

Problem 4.7 **Temperature dependence at low temperatures**

(i) Since $C_V = \gamma T$, we see

$$\left(\frac{\partial U}{\partial T}\right)_V = C_V \quad \Rightarrow \quad U(T) \simeq U(T=0) + \gamma \frac{T^2}{2} \ .$$

(ii) For the entropy, at constant volume use:

$$\mathrm{d}S = \frac{C_V}{T}\mathrm{d}T \ .$$

Thus:

$$S = \int_0^T \frac{C_V}{T}\mathrm{d}T \simeq \gamma T \ .$$

Here we assume $S(T=0) = 0$.

(iii) Since $F = U - TS$, we see

$$F(T) \simeq \underbrace{U(T=0) + \tfrac{1}{2}\gamma T^2}_{U(T)} - \underbrace{T\gamma T}_{S} = F - \tfrac{1}{2}\gamma T^2 \ .$$

(iv) Since $F + pV = G = \mu N$, and $p = 2U/(3V)$, we see that

$$\mu = \frac{G}{N} = \frac{F + pV}{N} = \frac{F + \frac{2}{3}U}{N} \ .$$

So

$$\begin{aligned}\mu &\simeq \underbrace{\frac{F(T=0) + \frac{2}{3}U(T=0)}{N}}_{\mu(T=0)} + \frac{T^2}{2N}(-\gamma + \tfrac{2}{3}\gamma) \\ &= \mu(T=0) - \frac{\gamma}{6N}T^2 \ .\end{aligned}$$

Problem 4.8 **Observability of Hawking radiation**

(i) T_{H} for the Sun is

$$\begin{aligned}T_{\mathrm{H}}^{\mathrm{S}} &= \frac{\hbar c^3}{8\pi k_{\mathrm{B}} G M_\odot} \\ &\simeq \frac{10^{-34} \times 3^3 \times 10^{24}}{24 \times 6.7 \times 10^{-11} \times 2 \times 10^{30} \times 1.4 \times 10^{-23}}\,\mathrm{K} \\ &\simeq \frac{10^{-10}}{20 \times 10^{-3}}\,\mathrm{K} \\ &= 5 \times 10^{-9}\,\mathrm{K} \ .\end{aligned}$$

Given the cosmic microwave background with a temperature of roughly 3 K, this would be unobservable.

(ii) To render $T_{\rm H}^{\rm s} > 3\,$K, and hence potentially visible, would need a mass $\sim 10^{-9} M_\odot$, this is like a largeish satellite such as lapetus, a moon of Saturn. Such bodies of course will not form stars – not massive enough for fusion, let alone large enough to have a black hole as their endpoint.

(iii) Tactic is to move all factors of M to the left hand side of the equation and group units appropriately on the rhs. Let $M_0 = M(t=0)$:

$$\begin{aligned}
\frac{{\rm d}M}{{\rm d}t} &= -4\pi \frac{1}{{\rm c}^2}\left(\frac{2GM}{{\rm c}^2}\right)^2 n_{\rm ch}\frac{\sigma}{k_{\rm B}^4}\left(\frac{\hbar {\rm c}^3}{8\pi GM}\right)^4 \\
\Rightarrow \quad \frac{1}{3}\frac{{\rm d}M^3}{{\rm d}t} &= -4\pi\frac{1}{{\rm c}^2} n_{\rm ch} 4 \frac{G^2}{{\rm c}^2}\frac{\pi^2}{60\hbar^3{\rm c}^2}\frac{\hbar^4}{(8\pi)^4 G^4} \\
\Rightarrow \quad \frac{{\rm d}M^3}{{\rm d}t} &= -\frac{3\times 16\pi^3}{60(8\pi)^4} n_{\rm ch}\frac{\hbar {\rm c}^4}{G^2} \\
&= -\frac{n_{\rm ch}}{5120\pi}\left(\frac{\hbar {\rm c}}{G}\right)^{3/2}\left(\frac{{\rm c}^5}{\hbar G}\right)^{1/2} \\
&= -\frac{n_{\rm ch}}{5120\pi} m_{\rm P}^3 t_{\rm P}^{-1} \\
\Rightarrow \quad M^3(t) - M_0^3 &= -\frac{n_{\rm ch}}{5120\pi} m_{\rm P}^3 \frac{t}{t_{\rm P}}
\end{aligned}$$

Thus the lifetime – when $M(\tau) = 0$ – is

$$\tau = \frac{n_{\rm ch}}{5120\pi}\frac{M_0^3}{m_{\rm P}^3}\, t_{\rm P}\ .$$

So upon substitution we find $\tau \sim 10^{68}\,$s, compared to the age of the Universe, $\sim 13\times 10^9\,$yr $\simeq 4\times 10^{17}\,$s. Big black holes will not be disappearing for a while, and the supermassive ones at the centre of galaxies with masses $\sim 10^9\, M_\odot$, have a life 10^{27} times longer.

(iv) We see that for black holes to be decaying in the current epoch, a factor of 10^{-51} is needed in the mass cubed factor, so

$$M_{\rm now} \sim 10^{-17} M_\odot \simeq 10^{13}\,{\rm kg}\ ,$$

roughly the mass of an asteroid or comet, eg Churyumov-Gerasimenko 67P, with a rather small Schwarzschild radius $R_{\rm s} \sim 10^{-14}\,$m, roughly ten times the size of a proton.